

Designed by Remus Teodorescu 2026

LECTURE 11.2

Dynamics and Energy

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Deep Learning for Engineering · 2026

MODEL · GRIP · JOULES



The same car, now as a dynamical system. A bicycle model with one parameter describes its motion, the tyres set a hard ceiling on its speed through every corner, and a lap costs joules in winding heat, rolling and the computer's draw. The lecture identifies the model from a coast-down and finds the cheapest lap with a network whose last layer cannot break the grip limit.

Learning Objectives

SECTIONS, IN ORDER

1. **The vehicle** — a bicycle model, and the grip limit it must respect

2. **The energy** — where a lap's joules go, and the cheapest lap

3. **Identify and optimise** — a coast-down, and the limit as a layer

Liu carries the inverse fit and the optimisation through a network. Every car number in the study notebook is a placeholder of the right order, marked as such; the on-car notebook measures your own, and the car has no encoder, so its speed comes from lap timing or from the camera.

READ, AND WHAT EACH ANSWERS

Liu ch. 4	what can a coast-down identify?
Liu ch. 5	how is a limit built in?
PyTorch	how is a cost differentiated?

The Kinematic Bicycle Model

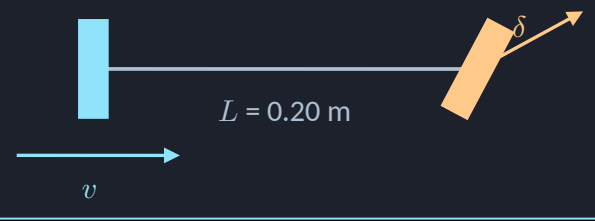
Kinematic bicycle model — the car reduced to two wheels: a speed and a steering angle in, position and heading out, with the wheelbase as its only parameter. Kinematic means no forces: it assumes the tyres never slide, so it holds exactly up to the grip limit and not an inch past it.

THREE LINES, ONE PARAMETER

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta, \qquad \dot{\theta} = \frac{v}{L}\tan\delta$$

v the speed, θ the heading, δ the steering angle, $L = 0.20$ m the wheelbase : the model's only parameter

THE CAR, REDUCED



two wheels stand for four

the rear drives, the front turns

one parameter: the wheelbase

BEFORE THE MODEL IS USED

the wheelbase	0.20 m, measured once
command to angle	not linear: calibrated
how	steady circles, radius taped
skip it	and every result is in doubt
suspension	real shocks: 'rigid' is false

The Pro has oil-filled shocks, so weight transfer is visible: the model is knowingly simplified.

The Grip Limit

Grip limit — the hard ceiling the tyres set: one budget of force, μmg, that cornering and braking share, so the speed through a bend of curvature κ cannot pass the square root of $\mu g/\kappa$. A penalty is a preference an optimiser may trade away; on a car, trading it away is a crash.

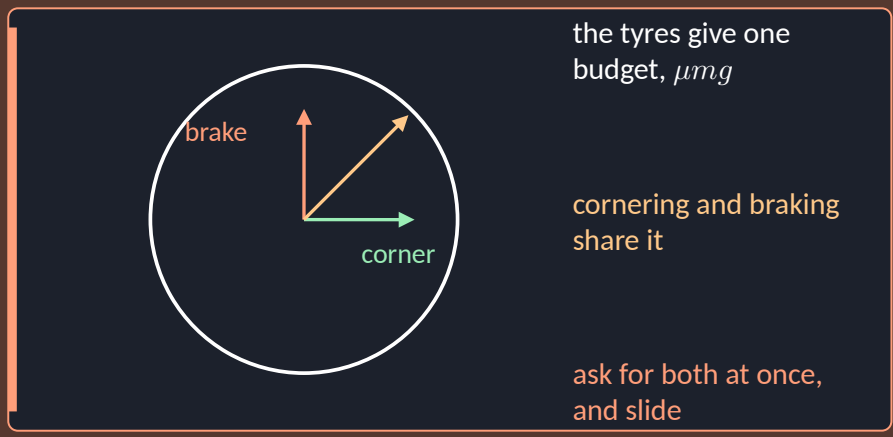
THE CEILING, POINT BY POINT

$$v(s) \le v_{lim}(s) = \min\left(\sqrt{\frac{\mu g}{\kappa(s)}},\ v_{\max}\right)$$

κ the curvature of the course, μ the tyre-floor friction, $v_{\max}$ the capped throttle's top speed

TALK · 2 MIN How would you explain, in your own words, the difference between a constraint in the loss and a constraint in the architecture? Then, in the drawing on the right, what happens to the cornering budget while the car brakes?

ONE BUDGET OF FORCE



soft: a penalty	traded when the loss prefers
hard: a layer	cannot be traded at all

L7.1's hard enforcement, with a crash as the stake.

Where the Joules Go

Lap energy — what the pack pays for one lap. The motor supplies rolling resistance, drag and acceleration; the current that takes is clamped at zero, because this drivetrain does not regenerate; and the power splits into winding heat, traction, and the computer's constant draw.

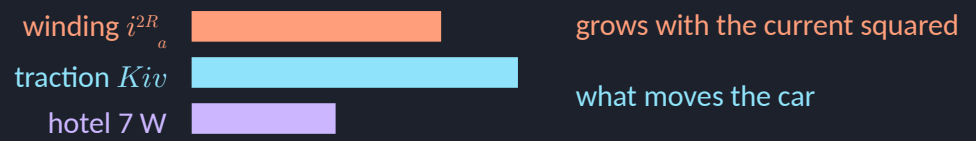
THE FORCE, THE CURRENT, THE POWER

$$F = c_{rr}\, mg + \frac{1}{2}\rho\, C_d A\, v^2 + m\, v\, \frac{dv}{ds}, \qquad i = \max\left(\frac{F}{K},\, 0\right)$$

$$P = i^2 R_a + K\, i\, v + P_{hotel}$$

c_{rr} rolling resistance, C_dA drag area, $m\,v\,dv/ds$ acceleration, K newtons per ampere; braking draws no current

i^2R_a winding heat, $K\,i\,v$ traction, $P_{hotel} = 7$ W: the Nano, the camera, the display, paid for the whole lap



THE PLACEHOLDER CAR, FLAGGED

	value
mass	1.05 kg
friction μ	0.65
rolling c_{rr}	0.02
motor K, R_a	1.0 N/A, 1.5 Ω
electronics	7 W, the whole lap
course	18 m, three corners

Placeholders of the right order, not measurements - problem.py marks every one. No result is your car's until the on-car notebook has replaced them. c_{rr} includes the transmission: the 4WD's drag is absorbed into it.

The Cheapest Lap

Lap cost — the energy of the lap plus the competition's price on its time, five joules a second. Driving slowly pays the electronics for longer; driving fast pays in winding heat and in the kinetic energy thrown away before each corner. So the cheapest lap is neither the slowest nor the fastest.

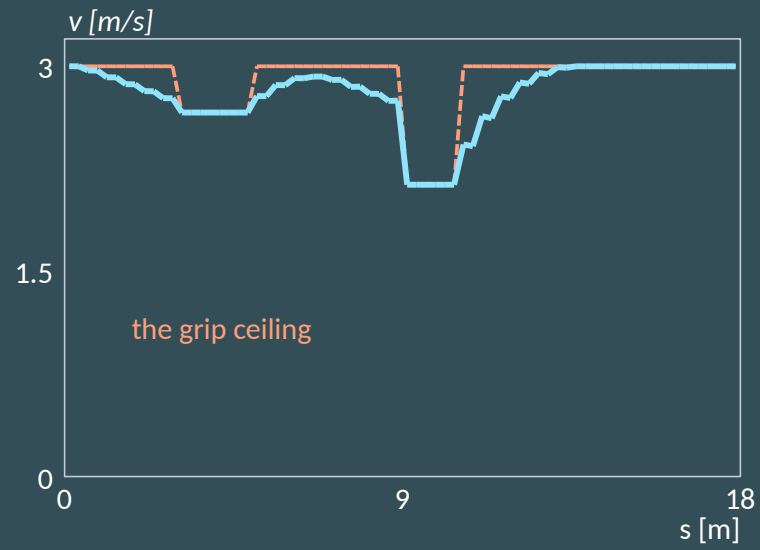
THE COST, AND WHAT THE OPTIMISER FOUND

$$J = E + w\,T = \oint \frac{P(v)}{v}\,ds \;+\; w \oint \frac{ds}{v}, \qquad w = 5 \text{ J/s}$$

E the lap's energy, *T* its time, *w* what the competition pays per second saved

On the placeholder car the classical answer is 55.60 J in 6.49 s, $J = 88.03$: with 7 W of electronics and 5 J/s, time dominates, so the best lap runs close to the ceiling and brakes smoothly into each corner - the smooth profile is the cheap one.

THE OPTIMISED SPEED, EX_11.2'S OWN



Dashed: the ceiling. Solid: the optimum, braking early and smoothly.

The Coast-Down

Coast-down — the identification run that costs nothing: cut the throttle and log the speed. No current flows, so only rolling and drag slow the car, and c_{rr} is read from the decay. A steady cruise cannot do this: there c_{rr} and R_a trade off, and the fit looks good anyway.

THE FREEWHEELING EQUATION, FITTED

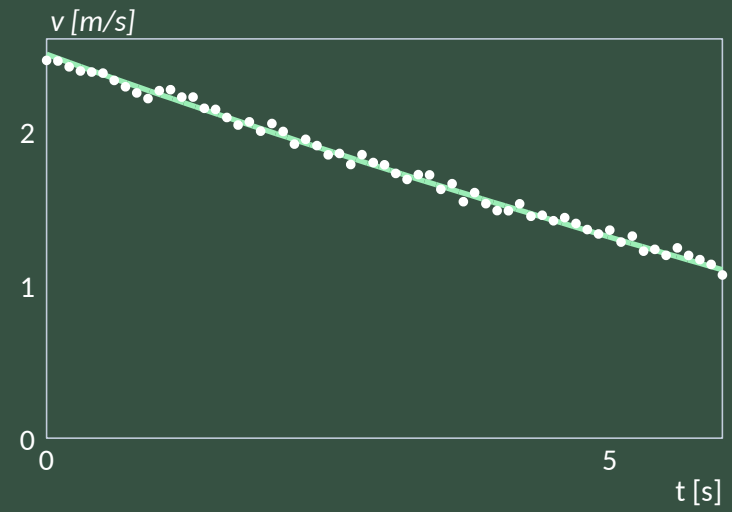
$$m\frac{dv}{dt} = -c_{rr}\,mg - \frac{1}{2}\rho\,C_dA\,v^2$$

$v(t)$ the speed of the freewheeling car : no current, so only rolling and drag slow it, and c_{rr} is read from the decay

THE PHYSICS-INFORMED FIT

```
c_rr = torch.nn.Parameter(torch.tensor(0.01))   # the coefficient, trained with
the fit
res = m * grad(v, t) + c_rr * m * g + 0.5 * rho * CdA * v**2   # the residual
L = res.pow(2).mean() + W_D * (v - log_v).pow(2).mean()        # physics + the
log
L.backward(); opt.step()                        # least squares agrees to 0.5 %
```

THE SYNTHETIC LOG, AND ITS FIT



Dots: the log, made with c_{rr} = 0.0200; least squares returns it, the physics-informed fit 0.0199. On the car the log is lap timing or the camera.

The Limit as a Layer

Bounded output layer — hard enforcement for an optimiser: a sigmoid squeezes the output and the local ceiling scales it, so the speed can approach the grip limit and never pass it, whatever the weights do. The cost is then minimised by gradient descent through the lap's physics.

THE SPEED PROFILE, BUILT SAFE

$$v(s) = v_{lim}(s)\,\sigma(\mathcal{N}(\cos ks,\ \sin ks)), \qquad k = 1 \dots 6$$

σ a sigmoid, between 0 and 1, $\mathcal{N}$ the network on six harmonics of the closed lap: v can approach the limit, never pass it

THE LAST LAYER

```
net = MLP(n_in=12, n_hidden=32, n_layers=3)     # six harmonics in: smooth, closed
v = v_lim * torch.sigmoid(net(h))               # can approach the limit, never pass it
```

TALK · 2 MIN How would you decide, in your own words, when an optimisation needs a network at all? Then, from the table on the right, what did the six harmonics buy that a plain sigmoid network did not?

AGAINST THE CLASSICAL OPTIMISER

	L-BFGS-B	network
energy	55.60 J	55.66 J
lap time	6.49 s	6.48 s
cost J	88.03 J	88.08 J
apart	—	0.05 %
over the ceiling	never	never, by build

288 bounded unknowns against a 3 × 32 network on six harmonics of the closed lap. The earlier notebook promised the network a few per cent worse; with the harmonics and L-BFGS it is 0.05 %. Here the classical optimiser is excellent - the network's case is built-in safety and reuse, not victory.

The Efficiency Trophy

Efficiency trophy — the block's competition: six cars, three clean laps each, two boards. Board one is the cheapest lap by measured cost; board two the best energy prediction, committed before the run - a model is judged by what it said in advance, not afterwards.

SIX CARS, TWO LEADERBOARDS

board one	the cheapest lap: least cost J over three clean laps
board two	the best prediction: committed before the run
clean	on the line, no cones, no hands
measured by	the car's own current sensor

WHAT EACH BOARD REWARDS

board one	driving and optimisation
board two	modelling and identification
one car	can win both
the write-up	says which numbers were yours

The study notebook's placeholder car scores nothing here: every parameter on the boards is one your team measured.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
the motor's own model	back-EMF, torque constant, winding	Waveshare wiki	where K and R_a come from
the INA219's arithmetic	shunt samples into a lap's joules	Waveshare wiki	the on-car measurement
the pack under load	voltage sag, internal resistance	Incropera ch. 1	long runs; the battery of L10.1
model predictive control	the plan re-solved as the car drives	Liu ch. 5	beyond a fixed profile
weight transfer	grip that moves between the axles	White ch. 2	braking into a corner

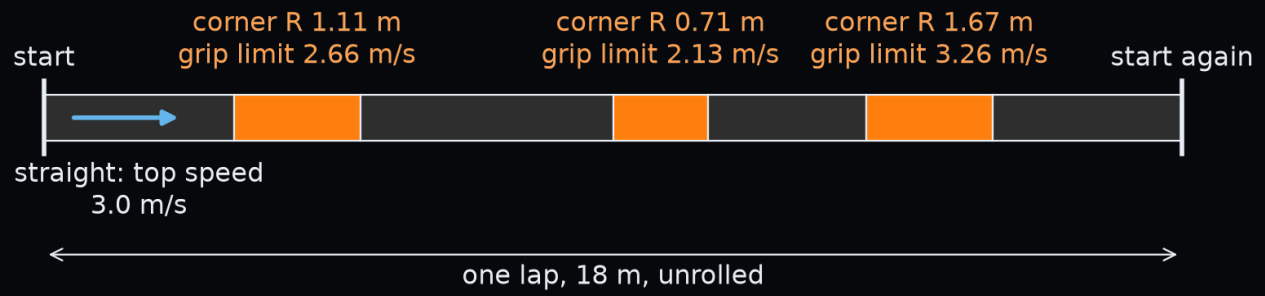
WHY THEY ARE LEFT OUT

Ex_11.2 optimises one speed profile on a known course, scored against its own classical optimiser, so none of them is needed to do it. The first two rows are the on-car notebook's territory; the last is why the bicycle model is knowingly simplified.

Exercise 11.2

ONE STUDY NOTEBOOK, AND THE CAR

- **1 – 3** — the car's data, the force, the power, the grip limit
- **4** — the classical optimiser, one bounded unknown per step
- **5** — the network with the limit as its last layer; the two compared
- **6** — the rolling coefficient from a coast-down, two ways
- **7 – 9** — the report; then Ex11.2_on_the_car: sensor, laps, boards



THE QUESTION EACH PART ANSWERS

1 – 3	what does a lap pay for?
4	what is the classical answer?
5	can a layer keep the car safe?
6	what does a coast-down tell?

REPORT QUESTION, TO CONCLUDE

On the car you commit an energy prediction before the scored laps, then measure. How far off was your prediction, which placeholder was most wrong, and what would you measure next to close the gap?

SUMMARY

Key Takeaways

VEHICLE MODEL	
what it is	a bicycle model, one parameter
function	commands to a path, up to grip
train, tune	wheelbase measured, map calibrated
vs the full car	no forces, no sliding, knowingly
used for	the lap's kinematics and ceiling

ENERGY	
what it is	winding heat, rolling, hotel load
function	the cost $J = E + wT$ of a lap
train, tune	c_{rr} from a coast-down
vs the slowest lap	time is paid too: a best inside
used for	the committed prediction, scored

Next — L12.1: the grid, where the network estimates what the meters cannot see.

Questions

1 – 2 · A NETWORK FOR THE CHEAPEST LAP

1 **How can a network find the speed profile of the cheapest lap while never passing the grip limit?**

The network takes the position along the lap, as six sine and cosine harmonics so that the end of the lap joins its start, and returns the speed there. The loss is the cost of the lap itself, the energy drawn from the battery plus the price on the lap time, differentiated by autograd. The grip limit is built into the last layer: a logistic function holds the speed between the stall speed and the local ceiling for any weights, where a penalty could be traded away for a lower cost. *Liu ch. 5; Ex_11.2 section 5*

2 **When does a network for the speed profile pay against the classical optimiser, and when does it not?**

In the exercise the classical optimiser, with one bounded speed per step of the lap, and the network agree within 0.05 per cent, and for one lap on a known course the classical optimiser is the right tool. The network pays when its answer is reused: it is a function of the position rather than a list of numbers, its grip limit holds by construction, and it can become a policy that reacts to what the camera sees. *Liu ch. 5; Ex_11.2 sections 4 – 5*

3 – 4 · THE ROLLING COEFFICIENT

3 **How can the rolling coefficient of the car be estimated with a network from a coast-down?**

The drive is cut and the speed is logged as it falls; with no current, only rolling resistance and air drag slow the car. A small network takes the time and returns the speed, and the rolling coefficient is one more trainable number, kept positive as an exponential. The loss is the residual of the freewheeling equation, computed by autograd, plus the misfit to the logged speed. On the synthetic log it returned the coefficient the log was made with, as least squares did. *Liu ch. 4; Ex_11.2 section 6*

4 **What must the measurements contain for the rolling coefficient to be identified, and when is it trusted?**

In a coast-down no current flows, so only rolling resistance and drag slow the car, and the decay of the speed separates them; in a steady cruise the rolling resistance and the winding resistance of the motor trade off, and a good fit can hide wrong values. The car has no encoder, so its speed comes from lap timing or the camera. Every car number in the notebook is a placeholder; the model is trusted only after it has predicted a lap that was then measured. *Liu ch. 4; Ex_11.2 section 6 and on the car*

NEXT

L12.1 — Power Grid Stability Estimation

The last block: the power grid, where a network proposes the state and the power flow equations judge it - with no true state to train on.

BEFORE THE NEXT SESSION

read	Liu ch. 5
run	Ex_11.2's study notebook, 1 to 6
bring	your lap cost, and the 0.05 %
for the lab	a working Ex_11.1 policy

Block eleven is finished: a car that sees, and a car that knows what a lap costs.